ADVANTEST®

V93000 Specifications

Pin Scale 1600 Digital Card

Version 8.6.0

Published on 2023-02-24

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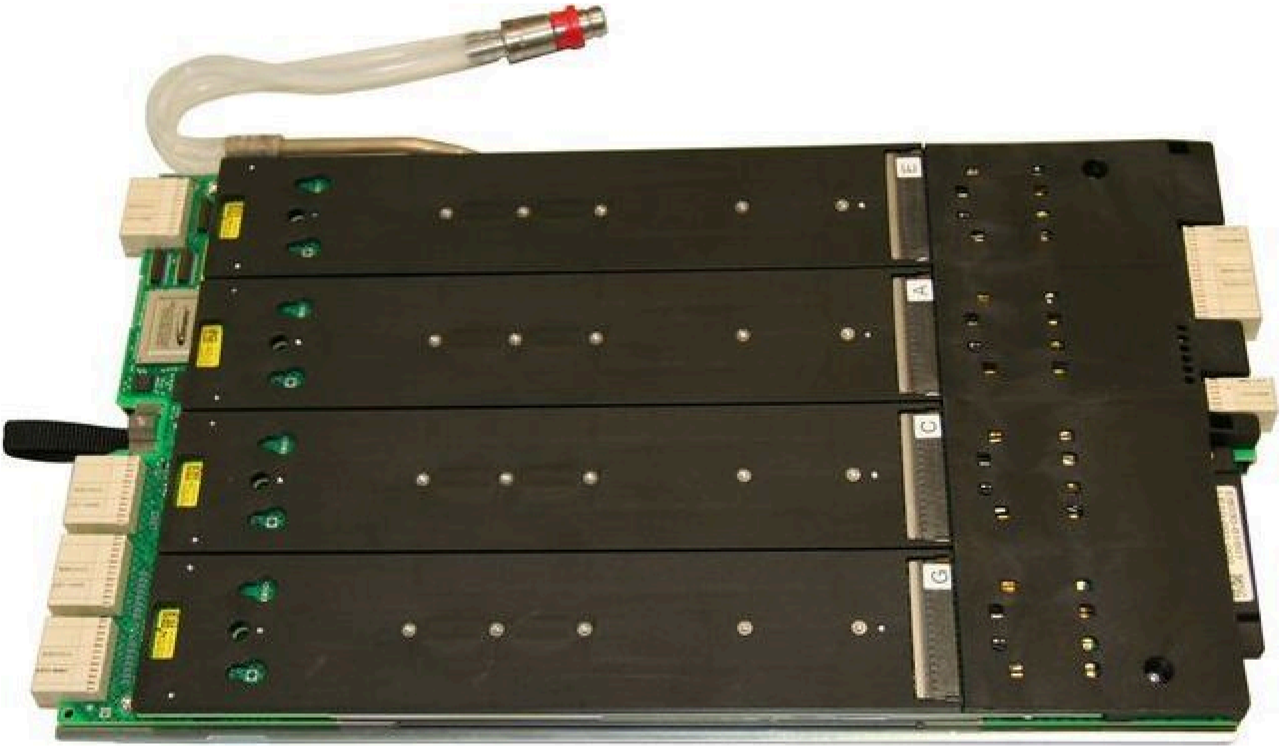
PS1600 specifications

Note	The specifications are subject to change without notice. Make sure to refer always to the latest revision found in the most recent SmarTest documentation (System Reference > Specifications).



Pin Scale 1600 Digital Card

Further information

For more information about the Advantest V93000 SOC Series, please visit *www.advantest.com*.

Contact information

For more information about the Advantest V93000 SOC cards, please contact your local Advantest sales representative.

This information is subject to change without notice.

PS1600 overview

Scope of specifications

Advantest specifies and verifies the specifications at the DUT interface pogo level. Using an adequate DUT board, *valid system calibration* and a fixture delay measurement (TDR), these specifications are valid at the device under test as well. Specifications describe warranted product performance. Characteristics are included as typical values to provide additional useful information by describing typical non-warranted performance.

Test Processor Per-Pin architecture

Channel count	128 channels per card
Parallel vector memory[1]	16 MByte (64 MVectors at x4 mode) standard. 32 MByte (128 MVectors at x4 mode) optional (N5867U). 64 MByte (256 MVectors at x4 mode) optional (N5868U). 112 MByte (448 MVectors at x4 mode) optional (N5869U). 224 MByte (896 MVectors at x4 mode) optional (N5870U) for the PS 1600B revision.
Maximum scan depth[2]	57 GVectors (at x4 mode) for PS1600A 114GVectors (at x4 mode) for PS1600B

Speed

Maximum I/O data rate at ≤ 1.8 V	1600 Mbps
Maximum I/O data rate at ≤ 3 V	1200 Mbps
Maximum clock rate at ≤ 1.8 V	800 MHz
Maximum clock rate at ≤ 3 V	600 MHz

Maximum external input voltage (compliance range)

Maximum external voltage range - for pins in standard voltage mode	 -3.0 V to +8.0 V -6.0 V to +16.0 V

[1] *Vector memory is used for pattern, setup data and result capture*
[2] *Maximum scan depth support total of 16 chains per instrument, 7.168GVectors max (drive only, single bit mode) or 3.584 GVectors in x4 mode for EACH chain. Total data volume per instrument ~ 57GVectors (in x4mode). For PS1600B: 14.336 GVectors max (drive only, single bit mode) or7.168 GVectors in x4 mode for EACH chain.*

Test programming and debugging

The SmarTest production test environment includes manufacturing test interfaces to operators, handling equipment, and factory networks. A graphical testflow editor links device tests into a production-ready test program, where the tests are set up via fill-in-the-blank test methods. User-specific tests are programmed with test methods in C++. Links are available for design-to-test conversion. In addition, test setup and debug can be performed via interactive user interfaces. Result analysis tools are available for error locations, timing behavior, analog waveforms, shmoo plots, pin margin tests and memory bit map.

Compatibility

This digital card is supported starting with SmarTest 7.1 It is a new member of the V93000 scalable platform expanding the digital card portfolio. It can be used with Pin Scale 9G, DC Scale, DPS, and Wave Scale RF cards. Configuration recommendations are available to optimize fleet utilization. .

Calibration and operation

Warm-up time	60 minutes
Basic maintenance period	6 months
Base calibration period (traceable calibration)	6 months
Calibration period[3]	3 months

Environmental

Operating	15°C to 30°C (59°F to 86°F) [4]
Specification warranted temperature	20°C to 30°C (68°F to 86°F)[4]
Maximum humidity at 30°C	< 70% R. H., non-condensing
Storage (without water)	-40°C to +70°C

[3] Valid at ambient temperature within ±2.5 K of calibration temperature. For temperature requirements during calibration please see "Maintenance Guide".

[4] For E8080E limited up to 25°C.

PS1600 Timing

PS1600 OTA and EPA

System OTA and EPA

Overall timing accuracy (t_{OTA})	< ±200 ps
Edge placement accuracy ($t_{EPA}=t_{OTA}/2$)	< ±100 ps

Per-board OTA and EPA

Overall timing accuracy per board (t_{OTA})	< ±160 ps
Edge placement accuracy per board ($t_{EPA}=t_{OTA}/2$)	< ±80 ps

Edge placement

Edge placement resolution	1 ps
Edge placement range[5]	-4 to 32 sequencer periods

PS1600 vector period

Vector period range	2.5 ns to 31250 ns
Vector period accuracy	±15 ppm of period setting
Vector period resolution	0.001 fs

PS1600 Clock Data Recovery (CDR)

Clock data recovery (CDR) operates in x4 mode, for example, operates up to 1600 Mbps.

Tracking range	±7.5 UI
Tracking max. frequency without clock forwarding	33.0 MHz for ±0.5 UI drift 2.2 MHz for ±7.5 UI drift
Tracking max frequency with clock forwarding	2.0 MHz for ±0.5 UI drift 139.0 kHz for ±7.5 UI drift

[5] Edge placement range maximal -6300 ns to +19000 ns. For sequencer period >752 ns the edge placement range is -4 to 12 sequencer periods

PS1600 driver

AC performance (Verification condition: half the noted voltage into 50 Ohm)

Maximum transition time (10 to 90%) at 1.0 V swing at 1.8 V swing at 3.0 V swing	 0.6 ns 0.7 ns 0.8 ns
Minimum pulse width at 1.0 V swing at 1.8 V swing at 3.0 V swing	 0.7 ns 0.8 ns 0.9 ns

DC performance (high, low, 3rd level, specified into open)

Level range	-1.5 V to 6.5 V
Level resolution	1 mV
Level accuracy	±5 mV
Minimum swing	100 mV[6]
Maximum swing	8 V
DC output current	±75 mA

Impedance characteristic

Source impedance	50 Ohm ± 2.5 Ohm

Z-clamp mode

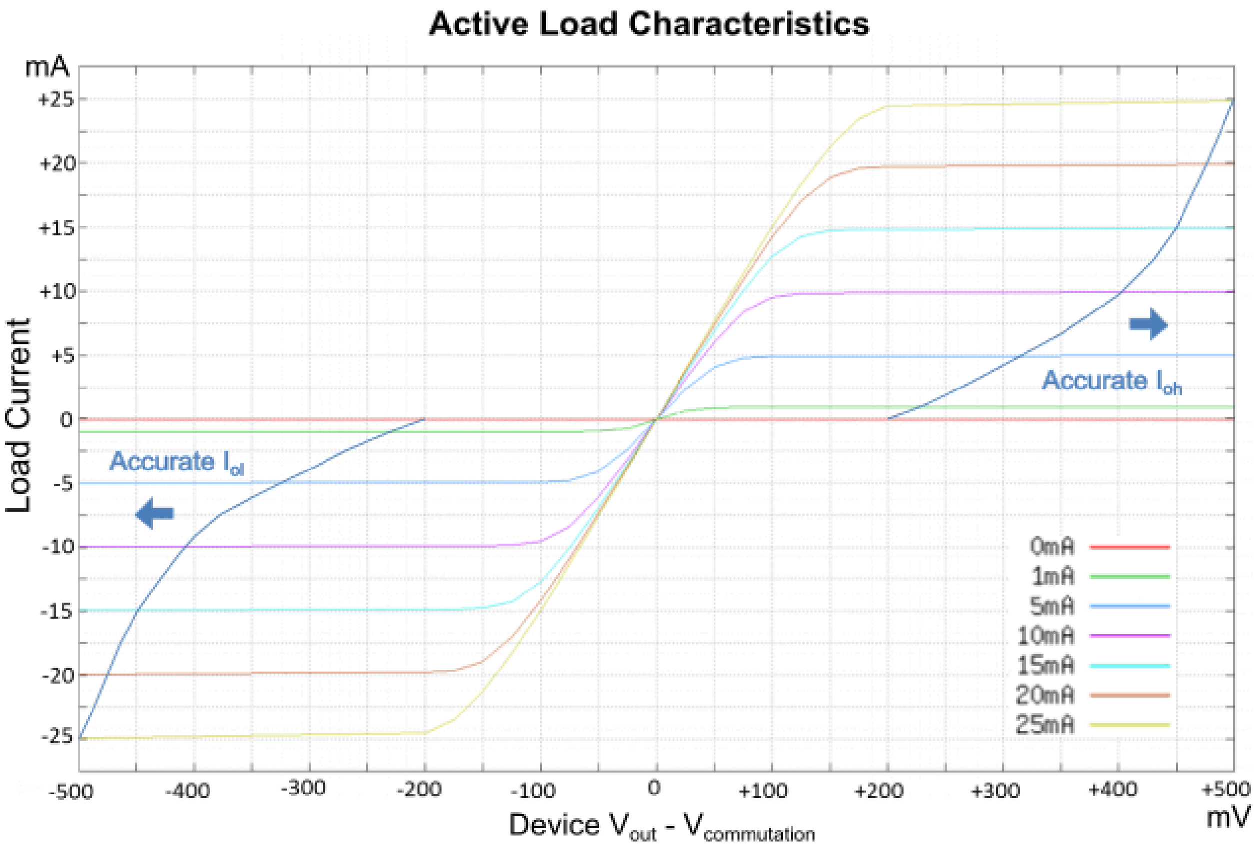
Voltage range	-2 to 7 V
Maximum voltage window	9 V
Voltage resolution	1 mV
Voltage accuracy	±100 mV[7]

[6] *Software allows programming to 0 mV swing, below the specified minimum swing*
[7] *calibrated at 900µA*

PS1600 programmable load

Currents (I_{oh}, I_{ol})	0 to 25 mA
Current resolution	12.5 µA
Current accuracy	±75 µA ±1% of max (I_{oh}, I_{ol})
Commutation voltage range (V_{com})	-1.0 V to 5.5 V[8]
Voltage resolution	1 mV
Voltage accuracy	±100 mV

Typical characteristics



Active load characteristics

[8] *-1.5 V to 6.5 V if I_{ol}, I_{oh} < 1 mA*

PS1600 receiver

AC performance

Minimum detectable pulse width (single ended and differential)	600 ps
Intrinsic transition time (10 to 90%)	350 ps, typical

DC performance - single ended

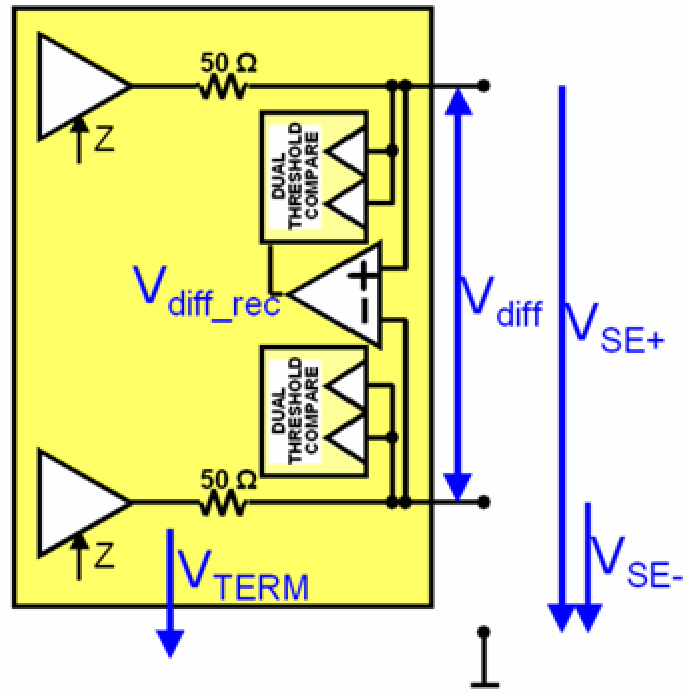
Input voltage and threshold range	-1.5 V to 6.5 V
Threshold voltage resolution	±1 mV
Threshold voltage accuracy	±5 mV
Pin leakage current	±2 uA

DC performance - differential

Input voltage range (V_{SE+}, V_{SE-})	-1.5 V to 6.5 V
Maximum differential input voltage (V_{diff})	±8 V
Differential receive voltage range ($V_{diff\_rec} = V_{SE+} - V_{SE-}$)	-1.0V to 1.0V
Threshold voltage resolution	±1 mV
Threshold voltage accuracy	±5 mV
Pin leakage current	±2 uA

Termination

Single ended / center tap differential impedance	50 Ohm ± 2.5 Ohm
Termination level center tap range	-1.5 V to 6.5 V
Termination voltage resolution	±1 mV
Termination voltage accuracy	±5 mV
Switchable pull-down resistor impedance	14.3 kOhm ± 10%


50 Ω
Z
DUAL THRESHOLD COMPARE
V diff_rec
+
-
V diff
V SE+
DUAL THRESHOLD COMPARE
50 Ω
Z
V TERM
V SE-

Receiver

PS1600 Parametric Measurement Unit - PMU per pin

PS1600 Voltage force and measure

Maximum Voltage range dependent on selected current range

Current Range within range 1 (± 40 mA) within range 2..5 (≤ ± 1 mA)	Voltage Range -2.0 V to 5.75 V -2.0 V to 6.50 V

Voltage force

Voltage resolution	200 µV
Voltage accuracy	±(3 mV + I*R)[9]

Voltage measure (pass/fail and value/SA) using PPMU level comparators

Voltage resolution	1 mV
Voltage accuracy	±(5 mV + I*R) ± 0.5% of reading[9]

Value measure using ADCs through PPMU

Voltage resolution	75 µV
Voltage accuracy in range -2.0 V to 6.5 V [10]	±4 mV
Voltage accuracy in range 0.0 V to 3.3 V	±2 mV
Relative voltage accuracy in range 0.0 V to 3.3 V[11] [10]	±1 mV

Capacitive loading

Supported PPMU capacitive load	≤ 50 nF or ≥ 10 µF

PS1600 Current force and measure

The sustainable total load may not exceed 1600 mA for Pin Scale 1600 (128 channels per card) and 800 mA for Pin Scale 1600-ME (64 channels per card).

Current force

	Range	Resolution	Force accuracy
Range 1	±40 mA	20 µA	±50 µA ±0.2% of setting

[9] *I is the actual current, R is the wiring resistance of ≤ 0.5 Ohm*
[10] *Uses 16 averaging samples internally*
[11] *In two consecutive voltage value measurements with different force currents, the accuracy of the calculated delta voltage*

	Range	Resolution	Force accuracy
Range 2	±1 mA	0.5 µA	±5.0 µA ±0.5% of setting
Range 3	±100 µA	50 nA	±500 nA ±0.5% of setting
Range 4	±10 µA	5 nA	±100 nA ±0.5% of setting
Range 5	±2 µA	1 nA	±40 nA ±0.5% of setting

Current measure using BADC through PPMU

	Range	Resolution	Measure accuracy
Range 1	±40 mA	20 µA [12]	±50 µA ±0.5% of reading
Range 2	±1 mA	0.5 µA [12]	±1.25 µA ±0.5% of reading
Range 3	±100 µA	50 nA [13]	±200 nA ±0.5% of reading
Range 4	±10 µA	5 nA [14]	±50 nA ±0.5% of reading
Range 5	±2 µA	1 nA [15]	±10 nA ±0.5% of reading

Current measure (value/SA) using PPMU level comparators

	Range	Resolution	Measure typical accuracy
Range 1	±40 mA	20 µA	±200 µA
Range 2	±1 mA	0.5 µA	±20 µA
Range 3	±100 µA		not supported
Range 4	±10 µA		not supported
Range 5	±2 µA		not supported

Range 1 force current up to ±56 mA for voltages ≤ 3.3 V. Range 1 force accuracy applies.

Current force for relative measurement

Relative accuracy[16]	±20 µA ±0.3% of setting in range 1

[12] *Uses 4 averaging samples internally*
[13] *Uses 8 averaging samples internally*
[14] *Uses 32 averaging samples internally*
[15] *Uses 64 averaging samples internally*
[16] *In two consecutive voltage value measurements with different force currents, the accuracy of the calculated delta of the forced currents*

Voltage clamp (available in current force)

Voltage range	-2.0 V to 6.5 V
Voltage resolution	1 mV
Accuracy	±100 mV

PS1600 board ADC

The board ADC provides very accurate high impedance voltage measurements through each digital channel. A board ADC is shared among 16 channels.

Measurement in standard voltage mode

Voltage range	-3.0 V to 8.0 V
Voltage resolution	75 uV
Accuracy [17]	±1 mV
Relative accuracy between two consecutive measurements on the same channel or channels sharing the same board ADC	**Voltage range -3.0 V to 8.0 V** ± 500 uV
Input impedance	> 100 MOhm

Further specification values on measuring waveforms with the board ADC are provided in *dcVI measureWaveform actions*

[17] Uses 16 averaging samples internally

PS1600 Time Measurement Unit - TMU per pin

This topic covers time measurement unit performance data.

DC performance

Specifications of receiver apply (see *PS1600 receiver* on page 10).

AC performance

Maximum input frequency	800 MHz
Linearity error	±5 ps
Accuracy, single timestamp to ARM (time zero)	±35 ps
Accuracy, skew/propagation delay (between different pin resources)	2 * (EPA + 35 ps)
Intrinsic jitter (RMS)	5 ps
Resolution	1 ps

Other (Maximum number of simultaneously active TMUs: 128 shared within slots 1-4 and slots 5-8)

Maximum sampling rate[18]	50 Msps
Measurement mode	TMU measurements of single ended or differential compared signals

[18] *The maximum number of samples is limited by the available channel memory.*

PS1600 Digitizer

Each Pin Scale 1600 channel can be used as a Digitizer for both voltage and current supporting two clock mode options:

- `standard` mode:

  The measurement for each sample is triggered from the sequencer.
- `lowjitter` mode:

  The samples are recorded with the BADC by triggering the FPGA. Higher sample rates are possible.

Each use model can be applied to these measurement types:

- High impedance voltage measurement

  All specifications for static Board ADC measurements apply.
- Force current, measure voltage

  All specifications for PMU per pin apply.
- Force voltage, measure current

  All specifications for PMU per pin apply.

Only 1 out of 16 consecutive channels can be used as digitizer at a time.

	lowJitter mode	**standard mode**
Max. Sample Rate	250ksps	34ksps
Sample Averaging	N/A (always 1)	available
Sample Memory (per waveform)	4K	limited to vector memory
Sample Period Resolution	25ns	8*vector period resolution

PS1600 High-Precision PMU

The High-Precision PMU (HPPMU) will not be supported with SmarTest version 8 and higher.